EstateHub

Real Estate Listing & Communication Platform

Complete project documentation: purpose, user roles, all features, architecture, database, security, setup and administrator access.

Technologies	HTML5, CSS3, vanilla JavaScript, Supabase (PostgreSQL, Auth, Storage, Row Level Security)
Alternative version	PHP 8 + MySQL for XAMPP (full server-side version)
Type	University project - all demo data is fictional

1. Project overview

EstateHub is a web platform where property owners publish properties for sale or rent, potential clients browse listings and talk to the owners, and administrators moderate everything. Every listing must be approved by an administrator before it becomes public.

The project exists in three versions that share the same features and design:

Version	How it works	Best for
EstateHub-Supabase (recommended)	One index.html file connected to a Supabase cloud database with real user accounts, image storage and server-enforced permissions.	Live demo, presentation, working with several users at once.
EstateHub (offline)	One index.html that stores everything in the browser (localStorage). No internet or account needed.	Quick offline demo on one computer.
real-estate-platform (PHP)	Classic PHP 8 + MySQL application for XAMPP, with sessions, CSRF protection and secure uploads.	Assignments that require PHP/MySQL.

2. User roles

Role	What the user can do
Guest (not logged in)	Browse approved listings, search and filter, view details and photos, read About Us, send the general Contact Us form. Pressing "Contact Property Owner" redirects to login.
Client	Everything a guest can do, plus register/login, message property owners from the property page, read replies, view conversation history and unread indicators.
Property Owner	Register/login, personal dashboard, create / edit / delete own listings with multiple images, see listing status (Pending, Approved, Rejected) and the rejection reason, receive and reply to client messages.
Administrator	Separate admin dashboard: statistics, approve / reject / edit / delete any listing, manage users, handle Contact Us messages, view the activity log.

3. Features in detail

3.1 Public website

- **Homepage** with hero section, quick search form, latest listings and navigation.
- **Property listings** with filters: sale or rent, location, minimum and maximum price, property type, minimum square meters and number of bedrooms.
- **Sorting** by newest, lowest price or highest price, and **pagination** (9 properties per page).
- **Property details**: image gallery with thumbnails, price, type, area, bedrooms, bathrooms, floor, construction year, energy class, description, owner name and dates.
- **About Us** and **Contact Us** pages. The contact form is open to everyone, is validated, and is protected by a hidden honeypot field, a minimum fill time and a submission limit (3 per hour).
- A **404 page** for unknown addresses.

3.2 Accounts and authentication

- Registration with full name, email, password, confirmation, optional phone and account type (Owner or Client). Admin accounts can **never** be created from the public form.
- Login and logout with sessions managed by Supabase Auth; passwords are never stored in plain text.
- Password rules: at least 8 characters with a letter and a number. Email addresses must be unique.
- Role-based access: every protected page and every database operation checks the user role.
- Deactivated accounts cannot log in or appear on the public site.

3.3 Property owner area

- **Dashboard** listing all own properties with status badges.
- **Add property**: title, description, sale/rent, type (apartment, house, maisonette, land, commercial, other), price in EUR, location, address, square meters, bedrooms, bathrooms, floor, construction year (optional), energy class (optional), and images.
- **Images**: JPG, PNG or WEBP up to 5 MB each; images are resized in the browser; choose the main image or remove images when editing.
- **Statuses**: new listings start as Pending. Editing an approved or rejected listing sends it back to Pending for a new approval.
- **Rejected listings** show the administrator's reason so the owner can fix and resubmit.
- An owner can never edit or delete another owner's listing.

3.4 Messaging between clients and owners

- The "Contact Property Owner" button opens a form with subject and message, linked to the specific property.
- One conversation exists per client and property. New messages are added to the same conversation.

- Owner and client both have an inbox with unread counters, a conversation page in chronological order with sender names and timestamps, and a reply form.
- Opening a conversation marks the other person's messages as read.
- Users can only open conversations in which they participate. Owners cannot message themselves.
- The inbox and conversation pages refresh automatically every 10 seconds.

3.5 Administrator panel

- **Dashboard statistics**: total users, owners, clients, properties, pending, approved, rejected properties and contact messages.
- **Manage properties**: filter by status; Approve, Reject (a reason is required), Edit and Delete any listing; open any listing in preview mode.
- **Manage users**: search, view role and status, activate or deactivate accounts, delete users with a confirmation dialog.
- **Contact messages**: read sender details and message, set status New / In Progress / Resolved, delete messages.
- **Activity log**: records property approvals, rejections, edits and deletions, user activation, deactivation and deletion, and contact message changes, with date and description.

4. Architecture and database (Supabase version)

The application is a single-page web app written in plain JavaScript. It talks directly to Supabase through the official JavaScript library. There is no custom server: security is enforced inside the database with Row Level Security (RLS) policies.

Table	Purpose
profiles	One row per user (name, email, phone, role admin/owner/client, active flag). Created automatically by a database trigger when someone registers; the trigger only allows the roles owner or client.
properties	Listings with all property fields, status (pending/approved/rejected), rejection reason and an array of image URLs.
conversations	One conversation per client and property (unique pair), with subject and last-update time.
messages	Messages of a conversation with sender, text and read flag.
contacts	Messages from the general Contact Us form with status new / in_progress / resolved.
logs	Administrator activity log.
pub_profiles (view)	Public names and roles without emails or phone numbers, used to show owner names.
Storage bucket property-images	Public bucket for property photos; users can only upload into a folder named with their own user id.

Main database rules (RLS)

- Visitors only see properties with status Approved whose owner is active. Owners also see their own listings; admins see everything.
- Owners can create listings only as Pending and can only change their own listings back to Pending - they can never approve them.
- Only administrators can approve, reject, manage users, read contact messages and read the activity log.
- Clients can start a conversation only for an approved property of another user. Messages can only be read and written by the two participants.
- Anyone, including guests, can insert a Contact Us message, but only admins can read it.

5. Security

- Passwords are hashed and managed by Supabase Auth.
- Permissions are enforced by the database (RLS), not only by the user interface.
- All user text is escaped before it is displayed (protection against XSS).
- Public registration can never create an administrator.
- File uploads are limited to JPG/PNG/WEBP images up to 5 MB.

- Only the public (publishable) Supabase key is used in the web page; the secret service key is never used.
- The PHP version additionally uses PDO prepared statements, CSRF tokens, session ID regeneration and login throttling.

6. Setup instructions (Supabase version)

- In the Supabase dashboard open **SQL Editor**, paste the file **supabase.sql** and press Run. If tables already exist, drop them first (the cleanup script was provided in the chat).
- In **Authentication - Providers - Email** turn off "Confirm email" so new users can log in immediately.
- Open **index.html** in a browser (internet connection required for the Supabase library).
- Register your own account, then promote it to administrator with one SQL command: `update profiles set role='admin' where email='YOUR_EMAIL';`
- Register an Owner and a Client with other email addresses to test listings and messages.

7. Administrator access

Item	Value
Application	EstateHub-Supabase / index.html
Admin email	dimossargentis2@gmail.com
Admin password	abcdefg12345
Supabase project	hebjfwrgaaxdqhmrmcyx

Important: this is a real account on a real Supabase project. Because this document contains the password in plain text, keep the PDF private. Before showing the project to others, change the password to a stronger one (Supabase - Authentication - Users, or by logging in and resetting it).

8. Suggested test scenario

- As a guest: browse listings, use filters and sorting, open a property, send the Contact Us form.
- Register as Owner and create a listing with images - it appears as Pending and is not public.
- Log in as admin: open Pending, approve one listing and reject another with a reason; check the Activity log.
- As the owner: see the approved status and the rejection reason; edit the rejected listing and see it return to Pending.
- Register as Client, open an approved property, press "Contact Property Owner" and send a message.
- As the owner: open the inbox, read and reply; as the client: see the reply and the unread counter.
- As admin: deactivate a user, change the status of a contact message, delete a listing.

9. Known limitations

- The Contact Us rate limit and honeypot run in the browser; a determined attacker could bypass them. A production system would add server-side checks (for example an Edge Function).
- Deleting a user removes the profile and all related data, but the login record remains in Supabase Authentication and can be deleted there.

- Images removed while editing a property stay in storage and can be cleaned up manually.
- Profile editing and "featured" listings are available in the PHP version only.